

Protocol



Construction of RNA reference materials for improving the quantification of transcriptomic data

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Abstract

RNA reference materials and their corresponding reference datasets act as the ‘ground truth’ for the normalization of experimental values and are indispensable tools for reliably measuring intrinsically small differences in RNA-sequencing data, such as those between molecular subtypes of diseases in clinical samples. However, the variability in ‘absolute’ expression profiles measured across different batches, methods or platforms limits the use of conventional RNA reference datasets. We recently proposed a ratio-based method for constructing reference datasets. The ratio for a gene is defined as the normalized expression levels between two sample groups and produces more reliable values than the ‘absolute’ values obtained across diverse transcriptomic technologies and batches. Our gene ratios have been used for the successful generation of omics-wide reference datasets. Here, we describe a step-by-step process for establishing RNA reference materials and reference datasets, covering three stages: (1) reference materials, including material preparation, homogeneity testing and stability testing; (2) ratio-based reference datasets, including characterization, uncertainty estimation and orthogonal validation; and (3) applications, including definition of performance metrics, performing proficiency tests and diagnosing and correcting batch effects. This approach established the Quartet RNA reference materials and reference datasets (chinese-quartet.org) that have been approved as the first suite of nationally certified RNA reference materials by China’s State Administration for Market Regulation. The protocol can be utilized to establish and apply reference materials to improve RNA-sequencing data quality in diverse clinical settings. The procedure can be completed in 2 d and requires expertise in molecular biology and bioinformatics.

Key points

- A ratiometric approach for quantifying reference samples, which includes material preparation, homogeneity testing and stability testing. The reference datasets are characterized and orthogonally validated, and their applications include benchmarking, accuracy testing and correcting for batch effects.
- Alternative gold-standard profiling technologies, such as the quantitative PCR standards, are limited when wanting to compare data from different instruments or methods.

Key references

Yu, Y. et al. *Nat. Biotechnol.* **42**, 1118–1132 (2024): <https://doi.org/10.1038/s41587-023-01867-9>

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Wang, D. et al. *Nat. Commun.* **15**, 6167 (2024): <https://doi.org/10.1038/s41467-024-50420-y>

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Introduction

RNA sequencing (RNA-seq) is an indispensable tool for transcriptome-wide analysis of differential gene expression and has demonstrated profound utility in clinical applications, such as molecular subtyping of diseases^{1–5}. The reliability of transcriptomics data, specifically the intrabatch proficiency and cross-batch reproducibility, is required for the routine clinical utility of RNA-seq data. The challenges in using RNA-seq data are further amplified when attempting the integration of multiomics data, such as combining transcriptomic data with other types of omics data to gain a comprehensive understanding of and high-level insights into human diseases. Hence, ensuring the reliability of transcriptomic research becomes paramount.

RNA reference materials and their corresponding transcriptome-wide reference datasets are valuable tools for evaluating the reliability of transcriptomic data. Reference materials are well-characterized samples with known properties. They can benchmark and validate experimental protocols, platforms and analytical methods. In contrast, reference datasets are well-characterized values of reference materials and can serve as the ‘ground truth’ for assessing the performance of these elements.

Previous studies have shown that omics-wide reference materials and reference datasets enable improvement of the reproducibility and accuracy of clinical applications in cancer^{6–8} and genetic diseases^{9–11}, as well as facilitating the development of new high-throughput technologies¹². Recently, we established the Quartet multiomics reference materials derived from immortalized B-lymphoblastoid cell lines from four members of a monozygotic twin family, including DNA, RNA, proteins and metabolites^{13–19}. Among them, the Quartet RNA reference materials have been proven to be an important resource for objectively assessing data quality and improving the reliability of transcriptomic profiling, specifically within a clinical setting¹⁵. Using the Quartet RNA reference materials, a study organized by China’s National Center for Clinical Laboratories systematically evaluated the performance across ~50 laboratories, revealing profound variations in data quality²⁰. Moreover, with the help of the Quartet multiomics reference materials, we identified reference-free ‘absolute’ feature quantification as the root cause of irreproducibility in multiomics measurement and demonstrated the advantages of the ratio-based multiomics profiling with common reference materials, highlighting the necessity for standardized benchmarks in multiomics research to ensure consistent and reliable data across various scientific studies^{13,15}.

Although other reference materials exist, the complexity of clinical scenarios necessitates a comprehensive palette of RNA reference materials. The RNA expression patterns vary dramatically across different sample types, from patient-derived tissues to biofluids, influenced by tissue-specific characteristics, disease states and individual genetic variability. Considering this diversity, developing tailored RNA reference materials and datasets is crucial, enabling the accurate representation of RNA profiles pertinent to distinct clinical scenarios and facilitating the molecular subtyping of distinct diseases.

The reference datasets need to be carefully established by integrating data from multiple measurement technologies. Unfortunately, the incomparability of conventional ‘absolute’ expression profiles across different methods or platforms prevents meaningful data integration. In our previous work^{13,15}, we found that ratio-based expression profiles, defined as the ratio of expression levels between two sample groups for the same gene, demonstrated strong consistency across multiple transcriptomic technologies, including RNA-seq, microarray and quantitative PCR (qPCR)^{21,22}.

In the following procedures, we provide the steps to construct transcriptomics reference materials and reference datasets, covering three stages (Fig. 1): (1) reference materials, including material preparation, homogeneity testing and stability testing; (2) ratio-based reference datasets, including characterization, uncertainty estimation and validation; and (3) applications, including constructing performance metrics, performing proficiency test and diagnosing and correcting batch effects. Several steps in the procedure can be simplified according to the intended use of the reference materials (Table 1).

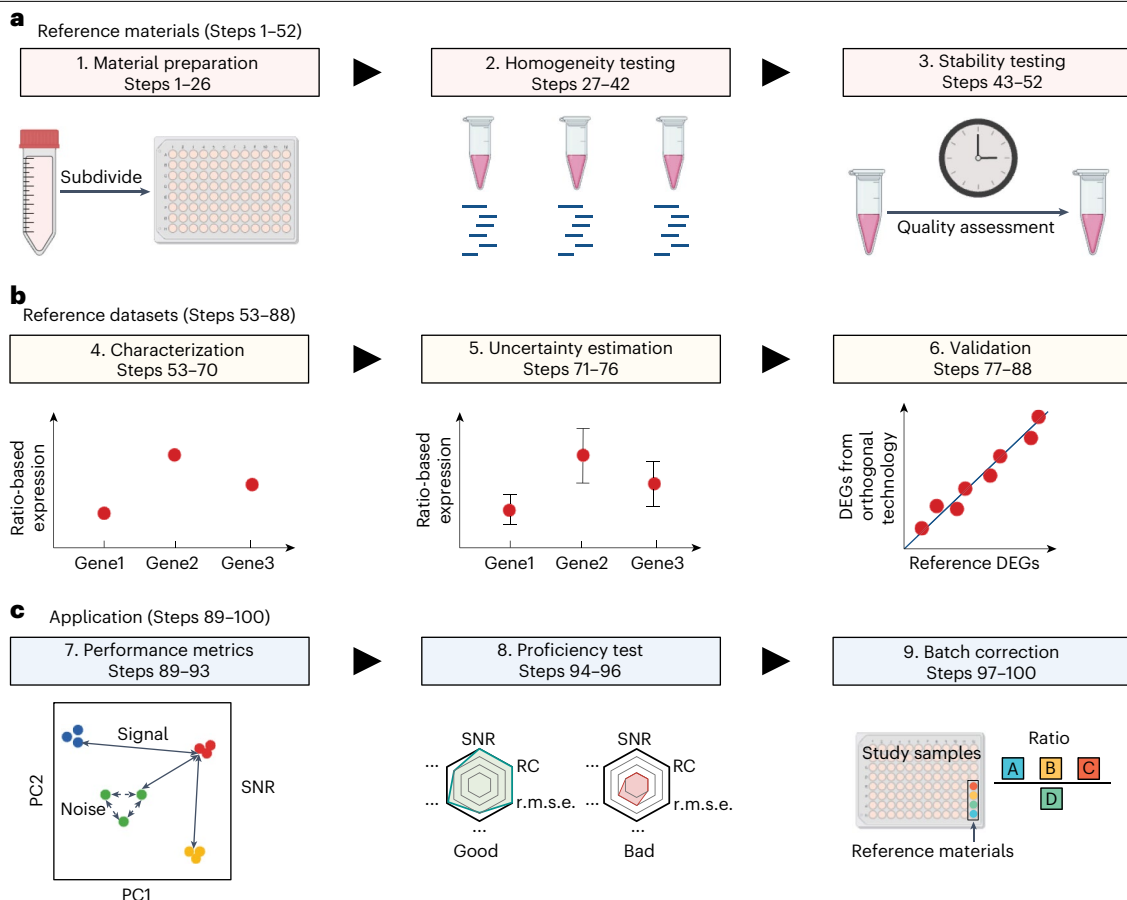


Fig. 1 | Overview of the protocol. **a–c.** Establishing RNA materials and reference datasets involves three stages: reference materials (**a**), ratio-based reference datasets (**b**) and applications (**c**). PC, principal component.

Development of the protocol

As part of the Quartet project for the quality control and data integration of multiomics profiling¹³, the RNA materials and reference datasets have been comprehensively constructed and validated. The Quartet RNA reference materials have been characterized as being high quality, homogeneous and stable and, therefore, can serve as an effective tool for the objective evaluation of data quality, thereby enhancing the reliability of transcriptomic analysis. Notably, the Quartet RNA reference materials have been approved by China's State Administration for Market Regulation, earning the distinction of being classified as the First Class of National Reference Materials.

Furthermore, several critical steps outlined in the procedure, such as characterization, performance measurement and batch-effect corrections based on reference materials, have been successfully adapted to other quantitative omics data types, including DNA methylation profiling, miRNA profiling, proteomics and metabolomics^{13–19}.

Applications of the method

The protocol can be applied to establish RNA reference materials derived from diverse sample types, such as cell lines, tissue samples, blood, plasma and so on. In addition, the protocol can be applied to diverse data types, such as alternative splicing in RNA-seq. Importantly, the protocol developed for constructing RNA reference materials and reference datasets can serve as a valuable foundation for other applications beyond RNA analysis, such as DNA methylation, miRNA, proteomics and metabolomics. Finally, the protocol may also work in establishing synthetic reference materials.

Table 1 | Recommended procedures for establishing reference materials

Stages	Procedures	Certified reference materials	Internal quality control samples
1	Reference materials	Candidate materials	●
2		Homogeneity testing	○
3		Stability testing	○
4	Reference datasets	Characterization	●
5		Uncertainty estimation	○
6		Validation	●
7	Construction of performance metrics		●

● Recommended procedures; ○ omitted procedures.

The protocol also outlines procedures for utilizing existing reference materials. These steps encompass proficiency testing, identification of batch effects and subsequent correction of such effects through the use of reference materials.

Comparison with other methods

One of the most widely used ways to characterize RNA reference datasets is using gold-standard profiling technologies, such as qPCR, which typically have a limited number of well-defined and characterized property values. For instance, the MicroArray Quality Control (MAQC) consortium profiled the expression levels of 1,044 genes in samples A and B using TaqMan qPCR assays as the reference datasets²². However, qPCR-based methods are unsuitable for characterizing omics reference materials that encompass many property values and reflect the complex and diverse nature of biological omics data²³. The qPCR-based reference datasets target specific genes, which means it cannot provide a comprehensive view of the entire transcriptome. Moreover, the results can be influenced by the efficiency of the primers used in qPCR experiments and lead to potential biases.

Experimental design

Material preparation

When selecting candidate reference materials, carefully consider sample types, the number of biological groups and the quantity of materials. First, choose sample types relevant to the research context, such as tissues, cell lines or biofluids, ensuring they represent the broader population or system of interest. Second, two or more biological groups should be used rather than relying on a single reference material. Repeated measurements of the same sample do not guarantee high sensitivity in identifying inherent biological differences (that is, signals) among various sample groups. Third, determine the appropriate amount of material to ensure it meets the specific needs of the study or trial. For example, in large cohort studies using reference materials for batch effect removal, if four groups of reference materials are produced, use one replicate of each group per batch of 96 libraries when using a 96-well plate. The required amount of reference materials will be the total number of study samples divided by 92 (96 – 4).

RNA-seq consideration

When performing RNA-seq, several important factors need to be considered for successful experimental design and data analysis. First, perform a proficiency test before performing large-scale RNA-seq experiments. It is essential to perform proficiency tests and method validation based on existing reference materials. Second, choose the library preparation approach(es), that is, poly-A selection or ribosomal-depletion approaches, based on the specific research question at hand. The two approaches create large differences in expression profiles because they are based on completely different principles in RNA preprocessing, namely enriching mRNA, and removing ribosomal RNA, respectively²⁴. Third, it is recommended to include at least three replicates per sample to ensure statistical power and robustness. Fourth, use the existing RNA reference materials as controls and conduct experiments

simultaneously with the candidate materials. Finally, consider the sequencing depth required for gene-level or transcript-level expression quantification.

Data preprocessing

Use well-established and widely adopted data analysis pipelines, such as HISAT–StringTie–Ballgown²⁵, because they have been thoroughly tested and assessed²⁶, are reliable and often come with extensive documentation and community support. Perform pipeline validation before obtaining expression profiles based on RNA-seq data. It is recommended that well-constructed RNA-seq data from existing reference materials, including built-in truth, be utilized.

Homogeneity testing

Homogeneity assessment ensures that the properties previously characterized for reference materials are evenly distributed across the packaging units. This assessment considers the homogeneity within a single unit and the consistency between different units. Clinically certified reference materials, often frozen liquids or lyophilized powder for resolution, typically exhibit minimal within-unit inhomogeneity. However, when aliquoting large pools of reference materials into thousands of units over several hours, variations can occur, leading to measurable differences in the analyte between units. Therefore, evaluating between-unit homogeneity becomes crucial.

Various experimental designs for between-unit homogeneity assessment are outlined in Fig. 2, following guidelines provided by ISO Guide 35 (2017)²⁷ and JJF-1343 (2012)²⁸. To assess between-unit homogeneity, multiple units are typically sampled from the entire lot of reference materials produced. The number of units selected for homogeneity assessment should be estimated on the basis of the lot size. Table 2 presents the typical numbers of units required for analysis. Additionally, homogeneity testing experiments should align with the primary purpose of developing reference materials. For instance, in our study, we utilized ratio-based gene expression profiles from RNA-seq to evaluate homogeneity, focusing on data quality in terms of gene expressions.

Stability testing

Stability testing aims to ensure that the value of the properties of the reference materials previously characterized remains unchanged over time. It evaluates the impact of environmental conditions during reference material shipment (short-term stability) and storage (long-term stability) by replicating similar conditions in the laboratory. Short-term stability should match the permissible transportation time, typically ~1 month. If specific transportation limits exist, stability duration can be adjusted. The testing usually involves three to five timepoints²⁸. Long-term stability testing determines the shelf life and storage duration of the reference material under expected environmental conditions, providing insights into functionality over time. The testing usually spans 6–24 months, with five to six sampling timepoints²⁸.

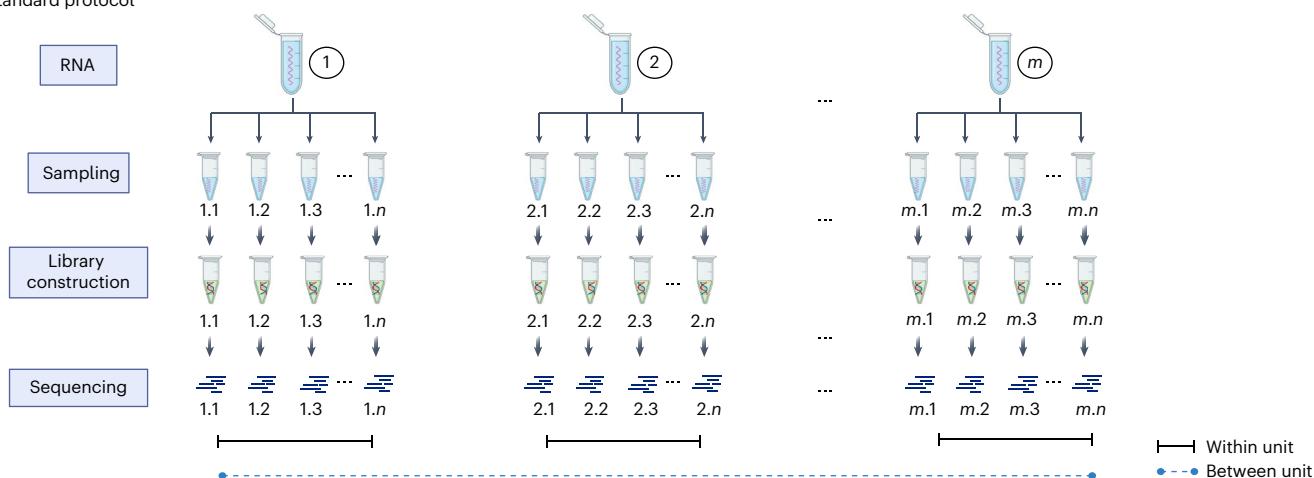
For performing stability testing, there are two main designs, that is, isochronous and classical (Fig. 3). The isochronous approach allows all measurements to be done in a single run. In contrast, the classical approach measures a sample as a function of time. Generally, an isochronous approach will lead to less uncertainty than a classical approach because the classical approach includes measurement instability under within-laboratory reproducibility conditions. On the other hand, a classical design is better suited for long-term stability testing and monitoring to provide data throughout the shelf life of reference materials because the isochronous approach is well suited for batch certifications at the end of the stability study.

Stability testing can be omitted when the total amount of reference materials is small (less than 100 units) and samples are stored under cryopreservation. The limited quantity and cryogenic storage conditions contribute to their inherent stability, reducing the need for extensive stability assessments. In this case, it remains essential to monitor their integrity during handling and use.

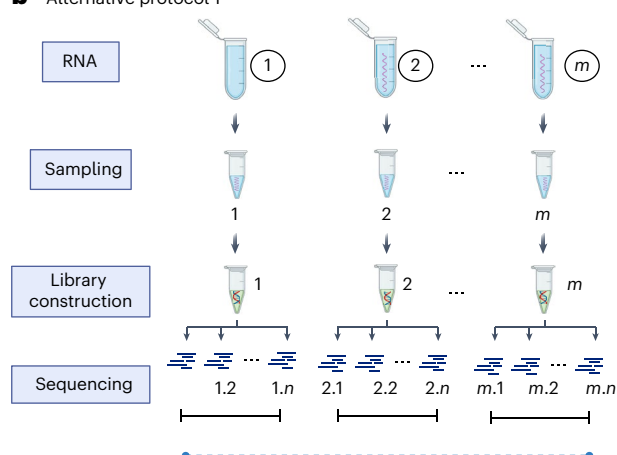
Characterization of reference datasets

When selecting a characterization method, it is essential to consider factors such as the type of reference material, usage requirements, relevant laboratory conditions, method performances

a Standard protocol



b Alternative protocol 1



c Alternative protocol 2

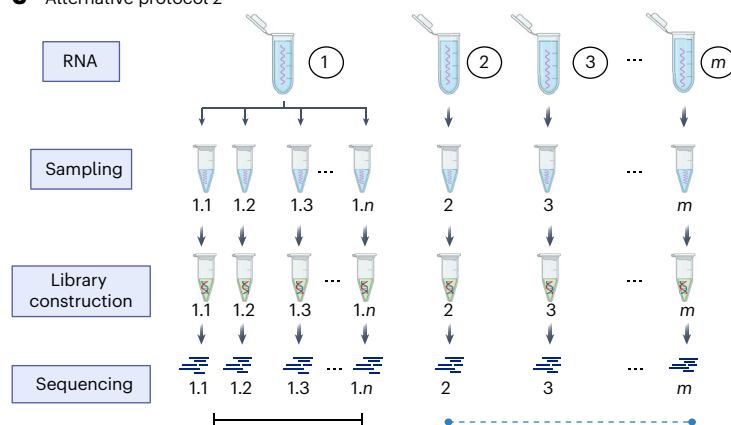


Fig. 2 | Workflows of between-unit homogeneity assessment. Three protocols are proposed for assessing homogeneity across different units. **a**, Standard protocol: each tube is sampled multiple times and, subsequently, multiple RNA-seq libraries and sequence runs are constructed. This approach comprehensively assesses homogeneity by considering variability across multiple replicates. **b**, Alternative protocol 1: each tube is sampled once, and a single RNA-seq library is constructed. Moreover, multiple sequencing runs are conducted on this single library. This approach balances efficiency

with homogeneity assessment while disregarding variations introduced during sampling and library construction steps. **c**, Alternative protocol 2: for tube 1, sampling, library preparation and sequencing are performed multiple times. By contrast, each remaining tube is sampled once, followed by library preparation and sequencing. This approach balances efficiency with homogeneity assessment while selectively focusing on specific tubes. In the chart, m represents the number of tube replicates, while n represents the number of sequencing replicates.

and the estimated uncertainty. Appropriate choices for characterization methods include collaborative efforts across multiple laboratories and/or technologies with distinct principles. Additionally, characterization can be performed on the basis of single or multiple methods with different principles. Our study uses collaborative efforts across multiple laboratories using multiple methods. Moreover, for characterization of transcriptome-wide reference materials, it is recommended to construct reference datasets at the ratio level.

Uncertainty estimation

Uncertainties of the reference datasets are essential for identifying each source of uncertainties and quantifying the uncertainty introduced by each source. The source of uncertainties can be classified into characterization uncertainties (u_{char}), sample inhomogeneities (between-bottle variation, u_{bb}) and instabilities (u_{s}), according to ISO Guide 35 (2017)²⁷, ISO/IEC Guide 93-3 (2008)²⁹ and JJF-1343 (2012)²⁸. These values are then combined to form the combined

Table 2 | Recommended number of units used for homogeneity testing

Total sample size	Minimum number of units
100 < <i>n</i> ≤ 200	11
200 < <i>n</i> ≤ 500	15
500 < <i>n</i> ≤ 1000	25
<i>n</i> > 1,000	30
<i>n</i> ≤ 500 (samples are expected to be homogeneous)	10
<i>n</i> > 500 (samples are expected to be homogeneous)	15

uncertainties (u_c) with a simple additive measurement model using an equal weight of the three uncertainty sources, as recommended^{27–29}. The expanded uncertainties (U) are further computed by multiplying u_c with an expansion factor.

Validation of reference datasets

Orthogonal methods that operate on distinct principles can be utilized to confirm the quantitative results of the reference datasets. Given that reference datasets are built using RNA-seq, it is advisable to employ either quantitative reverse transcription-polymerase chain reaction (RT-qPCR) or droplet digital PCR (ddPCR) for validation. These methods can measure the expression levels of a specific subset of genes in the reference datasets.

Expertise needed to implement the protocol

Implementing the protocol necessitates a range of expertise. Expertise in molecular biology techniques for handling biological samples, conducting next-generation library preparation and sequencing is essential. Moreover, a bioinformatician is necessary for processing the generated data and managing the pipeline. Additionally, researchers should also possess a basic understanding of statistical concepts that are related to homogeneity testing, stability assessment, uncertainty estimation and so on.

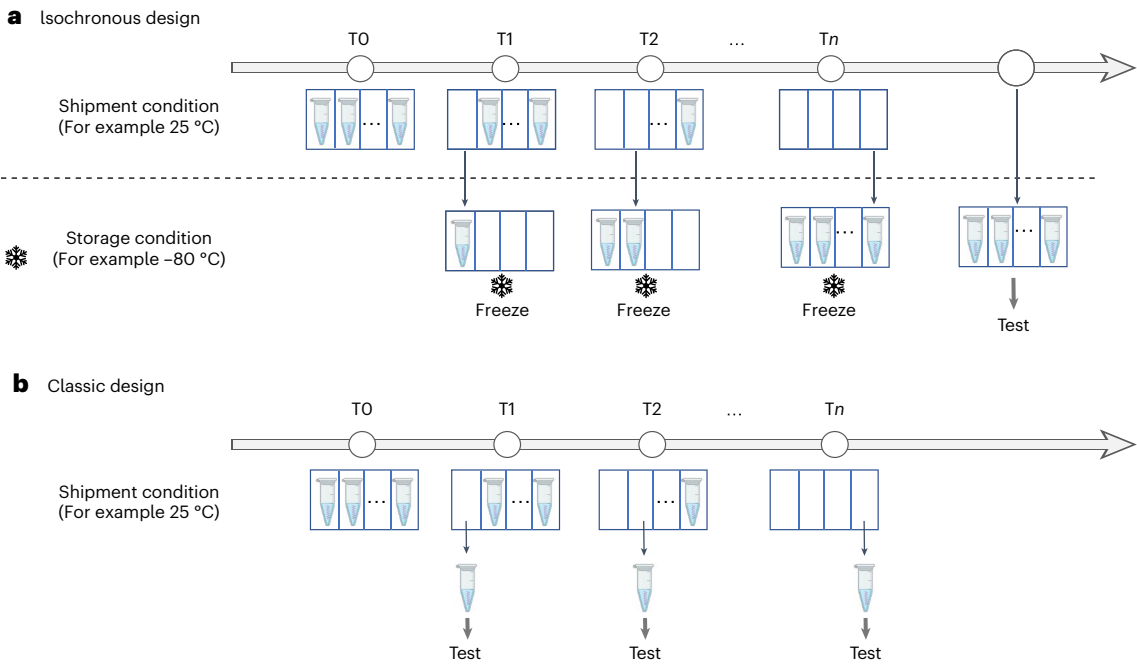


Fig. 3 | Scheme of isochronous design and classic design in stability testing. a,b. There are two main designs for performing stability testing, that is, isochronous (a) and classical (b). The isochronous design allows all

measurements to be done in a single run, while the classical design measures a sample as a function of time.

Limitations

The protocol has several limitations. First, the characterization of reference datasets relies on a single data analysis pipeline, which may introduce bias in transcriptomic quantification and establishment of the reference datasets. Although prior studies have compared the performance of different RNA-seq analysis tools and found overall good reproducibility for different tool combinations in terms of differential expression calls after proper filtering processes^{26,30,31}, it is essential to recognize that bioinformatics tools should undergo further evaluation for characterizing these reference datasets. Second, the reference datasets for biological samples may be incomplete, as genes can be removed when their expression values fall below the limit of quantification. Third, some genes may show inconsistent expression patterns when they can only be measured by specific extraction and analysis procedures. In this case, these genes will not be included in the reference datasets. Fourth, the reference datasets can be limited by the specific versions of genome references and gene annotations used.

Materials

Reagents

- Agilent RNA 6000 Nano Kit (Agilent, cat. no. 5067-1511)
- Buffer RWT (Qiagen, cat. no. 1067933)
- Chloroform (Sinopharm Chemical Reagent Co., cat. no. 10006818)
- Ethanol absolute (Sangon Biotech, cat. no. A500737)
- PrimeScript IV cDNA Synthesis Mix (Takara, cat. no. 6215A)
- Qsep DNA Standard Carthage S2 Kit (Bioptic, cat. no. C105101)
- Qubit 1× dsDNA High Sensitivity Assay Kit (Invitrogen, cat. no. Q33231)
- QX200 ddPCR EvaGreen Supermix (BioRad, cat. no. 1864033)
- Ribo-off Globin and ribosomal RNA (rRNA) Depletion Kit (Human/Mouse/Rat) (Vazyme, cat. no. N408)
- RNeasy max kit (Qiagen, cat. no. 75162)
- RNase-Free DNase Set (Qiagen, cat. no. 79254)
- RNase-Free water (Ambion, cat. no. AM9937)
- RNaseZap (Ambion, cat. no. AM9780)
- TRIzol Reagent (Invitrogen, cat. no. 15596018)
- UltraSYBR Mixture (CWBIO, cat. no. CW0957M)
- DNA Clean Beads (Vazyme, cat. no. N411)
- RNA Clean Beads (Vazyme, cat. no. N412)
- RNA Multiplex Oligos Set1- Set2 for Illumina (Vazyme, cat. no. N323/N324)
- Universal V6 RNA-seq Library Prep Kit for Illumina (Vazyme cat. no. NR604)

Equipment

- 0.5 mL Free Standing Screw Cap Microtube (QSP, cat. no. 516-Q)
- 8-Channel pipette (Eppendorf, cat. no. Research plus)
- Agilent 2100 System (Agilent, cat. no. G2991BA)
- Centrifuge (Eppendorf, cat. no. 5424R)
- Real-Time PCR System (BioRad, cat. no. CFX Connect)
- Microcentrifuge (Tiagen, cat. no. Cubee)
- NanoDrop spectrophotometer (Thermo Fisher, cat. no. ND-2000)
- NovaSeq sequencer (Illumina, cat. no. 6000)
- PCR tube (Axygen, cat. no. PCR-0108-LP-RT-W)
- Qsep analyzer (BIOptic, cat. no. 100 Advance)
- Qubit Fluorometer (Thermo Fisher, cat. no. 4.0)
- ddPCR system (BioRad, cat. no. QX200)
- Screw Cap Microtube with O Ring (QSP, cat. no. 525-MIXED-Q)
- The Multipette M4 (Eppendorf, cat. no. M4)

Protocol

- Thermocycler (BioRad, cat. no. C1000 Touch)
- Tips 0.1–10 μ L (QSP, cat. no. T104RLS-Q)
- Tips 1–200 μ L (QSP, cat. no. T090RLS-Q)
- Tips 100–1,250 μ L (QSP, cat. no. T112XLS-Q)
- Vazyme 0.2 mL magnet rack (Vazyme, cat. no. CM101)
- Vortex mixer (SI Vortex, cat. no. Genie2)

Software

- A Linux-based operating system
- Ballgown³², version 2.20.0 (<https://bioconductor.org/packages/release/bioc/html/ballgown.html>)
- Fastp³³, version 0.19.6 (<https://github.com/OpenGene/fastp>)
- FastQ Screen³⁴, version 0.15.3 (https://www.bioinformatics.babraham.ac.uk/projects/fastq_screen/)
- FastQC, version 0.12.1 (<https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>)
- HISAT³⁵, version 2.1.0 (<https://ccb.jhu.edu/software/hisat/index.shtml>)
- MultiQC³⁶, version 1.8 (<https://multiqc.info/>)
- Qualimap³⁷, version v2.3 (<http://qualimap.conesalab.org/>)
- R, version 3.6.5 or higher (<https://www.r-project.org/>) and Rstudio (<https://posit.co/downloads/>) or any other integrated development environment
- SAMtools³⁸, version 1.3.1 (<https://github.com/samtools/samtools>)
- StringTie³⁹, version 1.3.4 (<https://ccb.jhu.edu/software/stringtie/>)

Reagent setup

- RWT and RPE working solution: add the required volumes of ethanol (96–100%), as indicated on the bottle
- DNase I stock solution: inject 550 μ L RNase-free water to dissolve the lyophilized DNase I (~1,500 Kunitz units) using an RNase-free needle and syringe. Mix gently by inverting the vial
- ▲ **CRITICAL** Do not vortex the DNase I. DNase I is especially sensitive to physical denaturation.
- 80% (vol/vol) ethanol: prepare freshly with RNase-free water
- Dissolve each primer in RNase-free water to a final concentration of 100 μ M. Then dilute the primers to a final concentration of 10 μ M

Software setup

The software for the ‘RNA-seq data analysis’ section (Steps 31–39) is advised to be run on a Linux server. Detailed information and commands are provided in Supplementary Information. Users can follow this section using example data deposited in the Genome Sequence Archive (GSA) (accession number: HRA001859)⁴⁰.

For other types of analysis, such as homogeneity testing, stability testing, characterization and performance assessment, we recommend using RStudio with the R language. To facilitate organization, we encourage users to create a specific working directory (for example, ‘RNA_ref_data_analysis/’) to store example data and files generated during the analysis. Additionally, example data are available on GitHub (https://github.com/YingYu12345/Reference_standard_protocol) and Zenodo (<https://doi.org/10.5281/zenodo.13777366>)⁴¹.

Procedure

Material preparation

Purification of total RNA

● TIMING 3 h

1. Centrifuge the candidate material cells for 5 min at 400g and a temperature range of 15–25 °C. After centrifugation, discard the supernatant.

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2. For each batch of 1×10^8 cells, add TRIzol reagent to the collected cell pellet. Homogenize the sample thoroughly.
3. Add 2 mL of chloroform per 10 mL of TRIzol Reagent for lysis. Mix well to facilitate lysis. Then incubate the mixture for 5 min.
4. Centrifuge at 12,000g for 15 min at 4 °C.
5. Transfer 6 mL of the upper aqueous phase to a new tube.
6. Add 9 mL of 100% ethanol to the tube and mix thoroughly. Note that usually upper aqueous and 100% ethanol is 1:1.5 by volume.
7. Pipette up to 15 mL of the candidate material, including any precipitate that may have formed, into the RNeasy column in a 50 mL collection tube. Centrifuge the sample at 5,000g for 5 min at a temperature range of 15–25 °C. Discard the flow through and reuse the collection tube.
8. Repeat Step 7 using the remainder of the sample.
9. Add 7.5 mL buffer RWT to the RNeasy column. Centrifuge at 5,000g for 5 min at a temperature range of 15–25 °C. Discard the flow-through.
10. Prepare a DNase I incubation mix by adding 30 µL of DNase I stock solution to 210 µL of buffer RDD. Mix gently by inverting the tube.
11. Pipet 240 µL of DNase I incubation mix directly onto the RNeasy silica-gel membrane. Place the sample on the benchtop and incubate at a temperature range of 20–30 °C for 15 min.
12. Add 7.5 mL buffer RWT to the RNeasy column. Centrifuge at 5,000g for 5 min at a temperature range of 15–25 °C. Discard the flow through.
13. First wash: add 10 mL buffer RPE to the RNeasy column. Centrifuge at 5,000g for 2 min at a temperature range of 15–25 °C. Discard the flow through.
14. Second wash: add 10 mL buffer RPE to the RNeasy column. Centrifuge at 5,000g for 10 min at a temperature range of 15–25 °C to dry the RNeasy silica-gel membrane. Discard the flow through.
15. To elute, transfer the RNeasy column to a new 50 mL collection tube. After transferring, add 1 mL RNase-free water directly onto the RNeasy column. Let it stand for 1 min, and then centrifuge at 5,000g for 3 min at a temperature range of 15–25 °C.

Quality assessment of RNA samples

● TIMING 1–2 h

16. Use a NanoDrop ND-2000 spectrophotometer for RNA quantification and purity assessment, according to the manufacturer's instructions.
 - Turn on the NanoDrop ND-2000 spectrophotometer
 - Blank the instrument using nuclease-free water
 - Pipette 4 µL of each RNA sample onto the NanoDrop pedestal
 - Record the absorbance at 260 nm (A₂₆₀) to determine RNA concentration
 - Calculate RNA concentration using the appropriate conversion factor (an A₂₆₀ of 1.0 is 40 µg/mL for RNA)
 - Record the A₂₆₀/A₂₈₀ ratio to assess RNA purity (acceptable range: ~1.9–2.1)
 - Record the A₂₆₀/A₂₃₀ ratio to assess the absence of contaminants (acceptable range: ~2.0–2.2)
17. Use Agilent 2100 Bioanalyzer for RNA integrity assessment, according to the manufacturer's instructions.
 - Prepare RNA chips for the Agilent 2100 Bioanalyzer according to the manufacturer's instructions
 - Load 4 µL of each RNA sample onto the chip
 - Run the Bioanalyzer analysis
 - Examine the electropherogram to obtain the RNA integrity number (RIN). A RIN value of 9 or higher indicates intact RNA molecules with minimal degradation
18. Perform Steps 16–17 for three replicates of each RNA sample.
19. Calculate the average RNA concentration, A₂₆₀/A₂₈₀ ratio, A₂₆₀/A₂₃₀ and RNA integrity.
▲ **CRITICAL STEP** If an RNA sample meets these criteria, it is recognized as high quality and can proceed for gene-expression analysis, such as RNA-seq.

Subdividing RNA samples into aliquots

● TIMING 1–3 h

▲ **CRITICAL** After isolating RNA, it is common practice to divide the sample into smaller portions called aliquots. Aliquots are typically stored in labeled tubes or vials. In our study, an RNA aliquot with a volume of 10 μL is defined as a unit (vial) for storage, shipping and downstream analysis.

20. Mix the qualified RNA into a working aliquot for each RNA sample. Dilute the working aliquot to the determined concentration of 520 ng/ μL using RNase-free water. Thoroughly mix the solution.
21. Repeat Step 16 to measure the concentration of RNA. Perform three replicates for each RNA sample.
22. Dispense 10 μL of solution from Step 20 into each aliquot using the Multipette M4.
23. Store the aliquots of RNA in a -80°C freezer for long-term preservation.

■ **PAUSE POINT** The RNA aliquots can be stored at -80°C for several years without thawing.

Quality assessment of RNA aliquots based on multiple laboratories

● TIMING 2 h to 2 d (depending on the efficiency and speed of the external laboratories for independent assessment)

24. Randomly select three vials of aliquots to measure RNA concentration and integrity. Repeat Steps 16–19.
25. Randomly select six vials of aliquots and send them to two laboratories for independent quality assessment.
26. Calculate average RNA concentration, A260/A280 ratio, A260/A230 and RNA integrity from Steps 24–25.

Homogeneity testing (optional)

▲ **CRITICAL** If RNA materials are in the form of solutions or units are used at once, within-unit homogeneity might be negligible, while between-unit homogeneity testing shall be performed.

▲ **CRITICAL** Homogeneity testing should be conducted using data types similar to the intended use. For example, our study used ratio-based gene-expression profiles from RNA-seq to assess homogeneity to align with the primary purpose of developing reference materials.

Sample selection

● TIMING 1 h

27. Choose a workflow for between-unit homogeneity assessment. Three workflows are shown in Fig. 2. For our experiments, we utilized the alternative protocol 2 (Fig. 2c).
28. Randomly select a certain number of units for homogeneity testing. For a simple homogeneity study, select a minimum of 10 units.

Library preparation and sequencing

● TIMING 3 d

29. Process the candidate RNA reference materials using a standard library preparation kit based on the manufacturer's instructions. For our experiment, we used VAHTS Universal V6 RNA-seq Library Prep Kit for Illumina kit:
 - Use the appropriate cDNA synthesis buffer (generic or strand specific). Perform cDNA synthesis, end repair and dA-tailing in a single step without intermediate purification
 - Ligase adapters to the cDNA fragments. Amplify the ligated products using PCR primers. Incorporate heat-labile UDG to remove any remaining dUTP
 - Use magnetic bead-based size selection to obtain specific-length libraries. Customize library length according to your experimental requirements
 - Quality control using Agilent 2100 Bioanalyzer
30. Perform qPCR-based quantification, fragment analysis and sequencing:
 - Perform qPCR-based quantification to determine the concentration of each library using the library preparation kit for Illumina platforms
 - Determine sequencing depth based on your desired coverage for each library

- After preparing your libraries, perform paired-end sequencing using Illumina's constructs. For our experiments, we used the Illumina NovaSeq platform with 150 bp paired-end reads
- Obtain sequence files in fastq format, which are ready for downstream bioinformatic analysis
 - ▲ **CRITICAL** It is essential to perform library preparation and high-throughput sequencing in a randomized sample order to eliminate confounding factors such as experimental sample-processing order with sample group.

Alignment and gene quantification

● TIMING 6 h

▲ **CRITICAL** Detailed procedures are provided in Supplementary Information. All the command examples use the raw fastq data of sample D5_1 from batch R_BGI_L3_B1 in this protocol.

31. Use fastp to remove adapter sequences from fastq files. For a detailed description of fastp, visit <https://github.com/OpenGene/fastp>.
32. Perform RNA-seq alignment and gene-level quantification using HISAT, StringTie and Ballgown tools. Specifically, use HISAT to map the trimmed reads to the reference genome, such as GRCh38. Then use StringTie to assemble and estimate gene abundances for each sample. Next, use Ballgown to create read counts and fragments per kilobase of transcript per million mapped reads (FPKM) matrices. Pertea et al. provide a detailed protocol for these tools²⁵.
33. Add a value of 0.01 to the FPKM value of each gene and perform log₂ transformation.
34. Use expression profiles based on detected genes for further analysis. In our study, a gene is considered detectable (expressed) in a biological group within a batch if greater than or equal to three reads were mapped onto it in at least two of the three replicates.
 - ▲ **CRITICAL** Ballgown can process a single sample or merge the expression profiles of multiple samples.

Quality control analysis

● TIMING 2 h

35. Perform quality control on sequence data in fastq files using FastQC. For a detailed description of FastQC, visit <https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>.
36. Use FastQC on the fastq files generated by fastp to verify the success of adapter removal.
37. Use FastQ Screen on the fastq files generated by fastp to detect potential contamination with other species, junction primers and so on. For a detailed description of FastQ Screen, visit https://www.bioinformatics.babraham.ac.uk/projects/fastq_screen/.
38. Use Qualimap to calculate the quality of the reads mapped to different genomic regions. For a detailed description of Qualimap, visit <http://qualimap.conesalab.org/>.
 - ▲ **CRITICAL** Qualimap takes longer and requires more computational resources, whereas the RNA-seq quality control (QC) module may not run successfully due to version issues.
39. Use MultiQC to merge the QC results from FastQC, FastQ Screen and Qualimap. For a detailed description of MultiQC, visit <https://multiqc.info/>.

Compare variation between within-unit and between-unit groups

● TIMING 1 h

40. Perform statistical analysis (for example, analysis of variance) to compare the variation between the expression profiles of genes within-unit and those between-unit groups. The expression profiles are obtained from Step 34.
41. Calculate the adjusted *P* values with the 'fdr' method (for example, 'fdr' for Benjamini–Hochberg). A gene is considered homogeneous if the false discovery rate (FDR)-adjusted analysis of variance-based *P* > 0.05, indicating no significant difference between within-unit and between-unit groups.
 - ▲ **CRITICAL** The homogeneous analysis used here is relatively loose. Users can choose a more robust test of homogeneity.
42. Calculate the proportion of genes meeting the homogeneity criterion. If a majority of genes (for example, >95%) show homogeneity, the reference material is recognized as homogeneous. Run the script `homogeneity_assess.R` to execute Steps 40–42:

```
Rscript path-to/homogeneity_assess.R -i path-to/example_expr_homo_log2.csv -m path-to/metadata_homo.csv -o path-to/
```

▲ **CRITICAL** Two files will be obtained after successfully implementing the script, including: (1) the proportion of genes that meet the homogeneity criterion (`homogeneity_Ftest_summary_(date).csv`) and (2) the results of homogeneity of each gene under estimation (`homogeneity_Ftest_detail_(date).csv`).

Stability testing (optional)

▲ **CRITICAL** For small quantities of reference materials (less than 100 units) stored under cryopreservation, stability testing may be omitted. Nonetheless, monitoring their integrity during handling and use remains crucial.

Short-term stability

● **TIMING** varies

▲ **CRITICAL** If proper transport conditions are used to effectively reduce this uncertainty component, the effect of short-term stability can be neglected.

43. Select an approach for short-term stability assessment. Using an isochronous design (Fig. 3a) is recommended, where measurements are taken simultaneously.
44. Randomly select a certain number of units and expose them at room temperature (for example, 25 °C) and/or extreme temperature (for example, 40 °C) to represent shipment conditions for ~1 month. During each of the three to five timepoints within that month, place a few units (for example, three units) back into the storage condition (for example, 80 °C). Measure the RIN values and/or perform RNA-seq experiments after gathering all units for that month.

Stability testing based on RIN

● **TIMING** 4 h

45. After gathering all units for that month, repeat Step 17 to obtain the RIN values of each unit.
46. Calculate the ratio of RIN values using one reference material as the denominator. Perform regression analysis for each gene using the `lm()` function to estimate the observed slope and uncertainty of slope of RIN values across timepoints. If the observed slope is smaller than the uncertainty of the slope multiplied by the confidence level, the materials are stable. Run the script `stability_assess.R`:

```
Rscript path-to/stability_assess.R -i path-to/example_expr_stability_RIN.csv -m path-to/metadata_stability_RIN.csv -o path-to/
```

▲ **CRITICAL** One file named `stability_lm_detail_(date).csv` will be obtained after successfully implementing the script, illustrating the stability results of each gene or RIN values under estimation.

Stability testing based on RNA-seq

● **TIMING** varies

47. After gathering all units for that month, repeat Steps 29–39 to conduct RNA-seq experiments for each sample.
48. Repeat Step 46 to calculate the stability of each gene across timepoints. Calculate the proportion of genes meeting the stability criterion. The reference material is recognized as stable if a large proportion of genes is shown to be stable.

Long-term stability

● **TIMING** varies

49. Use the classical approach for long-term stability assessment (Fig. 3b). The classical design is advisable because measurements from the isochronous approach are typically available only at the end of the stability study.

50. Randomly select a certain number of units within a long-term period (usually 6–24 months) and perform measurements.
51. For each timepoint, measure RIN and/or perform RNA-seq of the selected samples.
▲ **CRITICAL** Conduct three replicates at each time point.
52. Repeat Steps 46 and 48 for long-term stability assessment.

Characterization

● **TIMING** varies

▲ **CRITICAL** The workflow of characterization is provided in Fig. 4.

RNA-seq from multiple protocols, platforms, laboratories and replicates

● **TIMING** varies (can be a few hours to a few days, depending on the instrument)

53. Send reference materials to multiple laboratories to conduct RNA-seq experiments using various protocols and sequencing platforms. For each batch of RNA-seq experiments, use three replicate units of each group reference material.
▲ **CRITICAL** Use existing reference materials as controls to generate RNA-seq data simultaneously.
54. Perform RNA-seq library construction and sequencing experiments following the standard procedures. Obtain the sequenced files in fastq format for downstream bioinformatic analysis. Repeat Steps 29–39 to conduct RNA-seq experiments for each sample.
▲ **CRITICAL** To avoid confounding factors related to experimental sample-processing order with sample groups, ensure that experiments within each batch are not performed in a fixed order by sample (for example, A-1, A-2, A-3, B-1, B-2 and B-3).

Data quality control based on expression profiles

● **TIMING** 2 h

55. Perform a principal component analysis (PCA) to visualize the major sources of variation of high-dimensional data in each batch. Use the `prcomp()` function in R.
56. Calculate the signal-to-noise ratio (SNR). The SNR is a measure often expressed in decibels that quantifies the strength of a signal relative to the background noise. For RNA-seq data, the ‘signal’ is the average distance that represents inherent ‘differences’ among various biological sample groups, whereas the ‘noise’ is the average distance among technical replicates within the same sample group in a space with reduced dimensionality. Generally, a lower SNR value suggests less discriminative power and vice versa. In our research, we have established the SNR threshold of 12. Run the script `SNR_qc.R`:

```
Rscript path-to/SNR_qc.R -i path-to/example_expr_multibatch_log2.csv -m path-to/metadata.csv -o path-to/
```

▲ **CRITICAL** Three files will be obtained after successfully implementing the script, including (1) the SNR value of the expression profiles of each batch (`SNR_perBatch_(date).csv`), (2) scatter plots of the PCA results with the SNR value for each batch (`SNR_perBatch_(date).pdf`) and (3) a scatter plot of the PCA results with SNR value across all batches (`SNR_allBatch_(date).pdf`).

◆ **TROUBLESHOOTING**

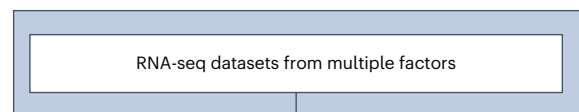
57. Calculate the Pearson correlations among technical replicates within the group of reference materials. Use the `cor()` function in R.
▲ **CRITICAL** The QC procedures, which are based on expression profiles, are conducted for each batch. Typically, a batch of data includes triplicates from each group of the reference materials being evaluated.

Identification of high-quality RNA-seq datasets

● **TIMING** 0.5 h

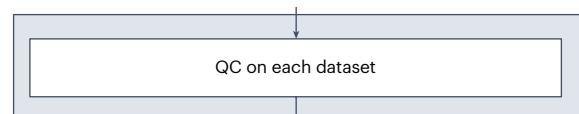
58. For each batch, perform a quality assessment from results generated from Steps 35–39 and 55–57. Identify high-quality RNA-seq datasets that meet all of the following criteria:

Steps 53–54: RNA-seq from multiple protocols, platforms, laboratories and replicates



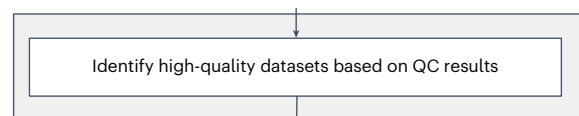
- Protocols, platforms, laboratories and replicates

Steps 55–57: data quality control based on expression profiles



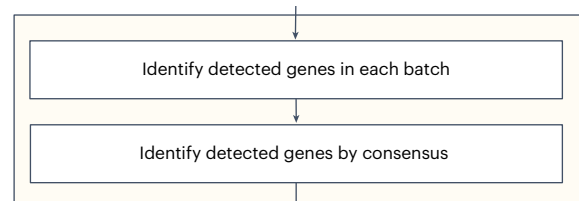
- PCA
- SNR
- MCC

Step 58: identification of high quality RNA-seq datasets



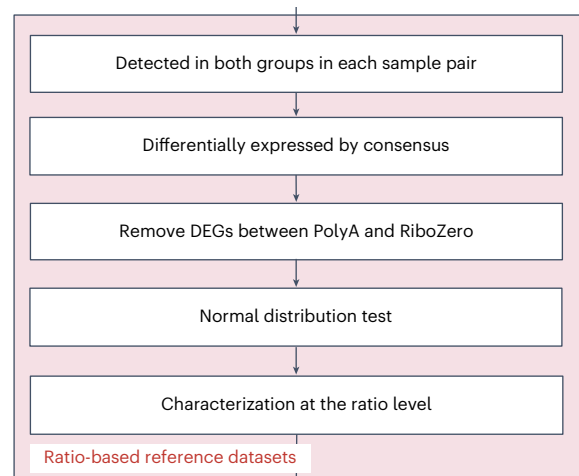
- Raw sequencing data
- Alignment data
- Expression data

Steps 59–61: identification of high-confidence detected genes



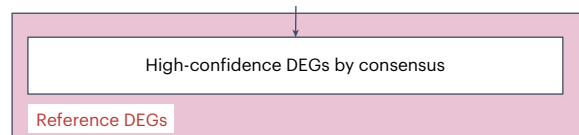
- ≥ 3 Counts in more than two replicates
- Detected across all batches

Steps 62–67: characterization of reference datasets at ratio level



- One sample pair includes two samples
- Limma-based $P < 0.05$ in all batches in any sample pair
- t-test $P < 0.05$ and $|\log_2 FC| > 1$
- Shapiro–Wilk test $P > 0.05$
- Geometric means across all batches

Steps 68–70: identification of reference DEGs



- Limma-based $P < 0.05$ and $|\log_2 FC| > 1$
- Select the most supported group

Fig. 4 | Workflow of characterization of reference datasets. Reference datasets can be constructed according to the following steps: (1) performing RNA-seq from multiple protocols, platforms, laboratories and replicates; (2) data quality control based on expression profiles; (3) identification of high-quality RNA-seq datasets; (4) identification of high-confidence detected genes; (5) characterization of reference datasets at ratio level; and (6) identification of reference DEGs.

- Sequencing read: number of reads, >10 million; read length, >35 bp; and the reads of adapter sequence $\leq 5\%$
- Q30 > 90%
- Contamination (for example, virus, bacteria or fungi) <2%
- rRNA or mitochondrial RNA (mtRNA) residuals <20%
- Mapping to human genome >90%
- Gene body (5'–3') distribution: -0.9–1.1
- Intronic region <10% across reads mapped to the human genome
- Correlation among technical replicates >0.9
- SNR >12

◆ TROUBLESHOOTING

Identification of high-confidence detected genes

● TIMING 0.5 h

59. Define expression criteria for considering a gene as expressed in a sample. A gene is considered expressed in a library in each batch if more than three reads were mapped to it in at least two of the three replicates.
60. Calculate the number of genes detected in each reference RNA sample group.
61. Conduct the detected genes across all samples. If a gene is detected beyond two of three replicates in a sample group of reference material, it is considered expressed in that sample group. Run the script `char_detect.R` to execute Steps 59–61 using the following command:

```
Rscript path-to/char_detect.R -i path-to/example_expr_multibatch_count.csv -m path-to/metadata.csv -o path-to/
```

▲ **CRITICAL** A file named `detect_genelist_(date).csv` will be generated upon successful script implementation. The file contains a list of genes classified as detected genes in each batch within each reference RNA sample group.

Characterization of reference datasets at the ratio level

● TIMING 0.5 h

62. Identify the detectable genes across the two groups of each sample pair.
63. Determine the ratio-based expressions based on detectable genes as follows:
 - (a) Obtain ratio-based expression data for each batch, calculated on a per-gene basis.
 - (b) Use $\log_2$ FPKM values to calculate these ratio-based expressions.
 - (c) For each gene, first compute the mean of the expression profiles of replicates of one group of the reference materials.
 - (d) Subtract this mean from the $\log_2$ FPKM values of that gene in each batch to obtain the ratio-based expressions.
64. To improve the reliability of the reference values, select genes that meet the threshold of $P < 0.05$ in each sample pair using the `limma` method⁴². Run the script `deg_limma.R` using the following command:

```
Rscript path-to/deg_limma.R -i path-to/example_expr_multibatch_count.csv -m path-to/metadata char.csv -o path-to/
```

▲ **CRITICAL** A file named `DEG_limma_(date).csv` will be generated upon successful script implementation. The file includes $\log_2$ -transformed fold change ($\log_2$ FC), averaged expression across both sample groups, t values, P values, FDR-adjusted P values and differentially expressed gene (DEG) classification in each sample pair.

65. Remove the genes that show significant differences between the two distinct protocols, PolyA and RiboZero, with a cutoff set at two-sided $P < 0.05$ from a Student's t -test and fold change (FC) ≥ 2 or ≤ 0.5 .

▲ **CRITICAL** The two protocols (PolyA and RiboZero) create large differences in expression profiles because they are based on completely different principles in RNA pre-processing, enriching mRNA and removing ribosomal RNA, respectively, before library construction.

66. Perform the Shapiro–Wilk test to check the normality of the ratio-based expressions. Ensure that data comply with normal distribution for statistical tests. Run the script `normality.R` using the following command:

```
Rscript path-to/normality.R -i path-to/DEG_limma_(date).csv -p path-to/batch_included.csv -o path-to/
```

▲ **CRITICAL** A file named `normality_genelist_(date).csv` will be generated upon successful script implementation. The file contains *P* values of the Shapiro–Wilk test based on the ratio-based expression of each gene in the reference datasets.

◆ **TROUBLESHOOTING**

67. Generate reference datasets at the ratio level for each pair of reference materials in the format of a mean by summarizing high-quality RNA-seq datasets. Run the script `char_ratio.R` using the following command:

```
Rscript path-to/char_ratio.R -i path-to/example_expr_multibatch_count.csv -m path-to/metadata.csv -g path-to/detect_genelist_(date).csv -d path-to/DEG_limma_(date).csv -o path-to/
```

▲ **CRITICAL** A file named `ref_expr_(date).csv` will be generated upon successful script implementation. The file contains ratio-based expression profiles of reference datasets, providing information on FC and median of limma-based *P* value for each gene.

Identification of reference DEGs

● **TIMING 0.5 h**

Establish high-confidence DEGs in the reference datasets, referred to as reference DEGs.

68. Identify DEGs in each batch. Our study uses the `limma`⁴² and `edgeR`⁴³ packages in R according to guidelines from the `limma` package. Specifically, `edgeR` is employed for data normalization based on a negative binomial model. At the same time, `limma` is used for identifying DEGs based on a linear model with expression profiles normalized via `edgeR`.
69. Within each batch, categorize genes as up- or downregulated, using a cutoff of a two-sided $P < 0.05$ from a `limma` test and a $FC \geq 2$ or ≤ 0.5 , respectively. Designate the remaining genes as non-DEGs.

▲ **CRITICAL** According to guidelines for microarray and RNA-seq data analysis from the MAQC and Sequencing Quality Control consortia⁴⁴, we used FC ranking with a nonstringent *P* value as a cutoff for identifying DEGs. This straightforward approach has been demonstrated to yield more reproducible and reliable lists of DEGs.

70. Identify reference DEGs and non-DEGs across high-quality batches. A gene is considered a reference DEG between two sample groups if it is consistently categorized as an up- or downregulated gene in more than a certain percentage of high-quality batches. In our study, we employed an arbitrary cutoff of 1/3. Then, run the script:

```
Rscript path-to/char_refDEG.R -d path-to/DEG_limma_(date).csv -r ref_expr_(date).csv -o path-to/
```

▲ **CRITICAL** A file named `RefData_DEGs_(date).csv` will be generated upon successful script implementation. This file includes a list of genes classified as reference DEGs for each sample pair. Additionally, it lists the average FC and median *P* values across high-quality batches for each gene.

◆ **TROUBLESHOOTING**

Uncertainty estimation

● TIMING 2 h

▲ **CRITICAL** Follow the guide to the expression of uncertainty in measurement method.

71. Characterization uncertainty refers to the uncertainty associated with the characterization of reference datasets. It is calculated on the basis of datasets from Step 58. Specifically, for each gene generated in Step 67, uncertainty is estimated using FCs ($\log_2$ transformed) from high-quality datasets. Run the script:

```
Rscript path-to/uchar.R -i path-to/DEG_limma.csv -p path-to/passbatch.csv -r path-to/ref_expr_(date).csv -o path-to/
```

▲ **CRITICAL** Upon successful script implementation, one file named Refdata_uchar_(date).csv will be generated, containing the characterization uncertainty of each gene in the reference datasets.

72. Inhomogeneity uncertainty arises when the material being measured is not uniform throughout. It is calculated using datasets from Step 34. Run the script:

```
Rscript path-to/ubb.R -i path-to/example_expr_homo_log2.csv -m path-to/metadata_homo.csv -r path-to/ref_expr_(date).csv -o path-to/
```

▲ **CRITICAL** Upon successful script implementation, one file named Refdata_ubb_(date).csv will be generated, containing the homogeneity uncertainty of each gene in the reference datasets.

73. Instability uncertainty accounts for variations over time due to environmental conditions or other factors. It can be calculated using RNA-seq datasets or RIN values to illustrate overall RNA integrity (Step 46). In our study, we employed linear regression profiles of RIN to represent overall instability. Run the script:

```
Rscript path-to/us.R -i path-to/example_expr_stability_RIN.csv -m path-to/metadata_stability_RIN.csv -r path-to/ref_expr_(date).csv -t R -o path-to/
```

▲ **CRITICAL** Upon successful script implementation, one file named Refdata_us_(date).csv will be generated, containing the instability uncertainty of each gene in the reference datasets.

74. Combined uncertainty accounts for all sources of uncertainty affecting a measurement, including characterization uncertainty (Step 71), inhomogeneity uncertainty (Step 72) and instability uncertainty (Step 73). Run the script:

```
Rscript path-to/uc.R -c Refdata_uchar_(date).csv -b Refdata_ubb_(date).csv -s Refdata_us_(date).csv -r path-to/ref_expr_(date).csv -o path-to/
```

▲ **CRITICAL** A file named Refdata_uc_(date).csv will be generated upon successful script implementation. These files contain the combined uncertainty of each gene in the reference datasets.

75. Extended uncertainty provides a confidence interval around the measured value. Apply a coverage factor (typically, $k = 2$ for a 95% confidence level). Multiply the combined uncertainty by k to obtain the extended uncertainty. Run the script:

```
Rscript path-to/U.R -d Refdata_uc_(date).csv -r path-to/ref_expr_(date).csv -k 2 -o path-to/
```

▲ **CRITICAL** A file named Refdata_U_(date).csv will be generated upon successful script implementation. These files contain the extended uncertainty of each gene in the reference datasets.

76. For reporting, express the final result as value $\pm$ extended uncertainty (for example, $Y = y \pm U$), where the value is determined in Step 67. Y represents the final format of the characteristic value, y represents the characteristic value and U represents the extended uncertainty.

Protocol

Orthogonal validation

RT-qPCR

●TIMING 3h

77. Randomly select a subset of genes with larger FCs in relative expression levels in the standard reference dataset and choose a reference gene (for example, *c1orf43*). Design the primers using the NCBI online software Primers-BLAST (<https://www.ncbi.nlm.nih.gov/tools/primer-blast/>).
78. Prepare the reverse transcription reaction mix described in the table below. Keep chilled on ice.

Reagent	Volume (μL)
5× PrimeScript IV cDNA Synthesis Mix	4
Random 6 mers (50 μM)	1
RNA template	1
RNase-free H ₂ O	14
Total	20

Mix thoroughly by pipetting ten times and then seal. Centrifuge gently (~300g for 1 min) to spin down the reaction buffer to the bottom of the tubes.

79. Transfer the tubes to a thermocycler and run the following program:

Temperature (°C)	Time (min)
30	10
42	15
95	5
4	Hold

80. Prepare the reverse transcription reaction mix described in the table below. Keep chilled on ice.

Reagent	Volume (μL)
2× UltraSYBR mixture	10
Forward primer (10 μM)	0.4
Reverse primer (10 μM)	0.4
Template DNA from Step 101	2
RNase-free H ₂ O	7.2
Total	20

Mix thoroughly by pipetting ten times and then seal. Centrifuge gently (~300g for 1 min) to spin down the reaction buffer to the bottom of the tubes.

81. Transfer the tubes to a thermocycler and run the following program:

Temperature (°C)	Time (s)	Cycles
Lid 105	On	
95	600	1
95	15	35–40
60	60	35–40
95	15	1
60	60	1
95	15	1
60	15	1
4	Hold	

Protocol

ddPCR

● TIMING 3h

82. Select the same genes as those validated by RT-qPCR and use the corresponding primers for ddPCR.
83. Prepare the reaction mix, as described in the table below. Keep chilled on ice.

Reagent	Volume (μL)
2× Evagreen Supermix	10
Forward primer	1
Reverse primer	1
Template DNA from Step 101	2
RNase-free H ₂ O	6
Total	20

Mix thoroughly by pipetting ten times and then seal. Centrifuge gently (~300g for 1 min) to spin down the reaction buffer to the bottom of the tubes.

84. Pipet the 20 μL reaction mixture into the sample wells of the corresponding droplet-generating card. Add 70 μL of droplet oil to the well. Transfer 40 μL of droplets to a 96-well plate for PCR reaction.
85. Transfer the well to a thermocycler and run the following program:

Temperature (°C)	Time (s)	Cycles
Lid 105	On	
95	300	1
95	30	35–40
60	60	35–40
4	300	1
90	300	1
4	Hold	

86. Transfer the well to the droplet reader for results.

Validation assessment

● TIMING 0.5 h

87. Calculate the correlation of reference datasets and validated datasets at the ratio level.
88. Generally, the consistency between the RT-qPCR or ddPCR and RNA-seq is ~0.8–1.

▲ **CRITICAL** Note that the level of the correlation coefficients depends on the level of the intrinsic biological differences between sample pairs under comparison. If the differences among the RNA reference materials are relatively small, the concordance between the reference datasets and the RT-qPCR or ddPCR data is also relatively small.

◆ TROUBLESHOOTING

Performance metrics

● TIMING 2h

▲ **CRITICAL** During data preparation step, the input files are processed. All the command examples in the R environment use the FPKM data (exprMat_log2FPKM.csv) from the high-quality batches in this protocol. For other samples, such as the data of other batches, change the name of the input and output files accordingly.

89. Use high-quality and low-quality datasets to construct reference-based performance metrics.
90. Calculate relative correlation (RC) with reference dataset metrics. RC is calculated based on the Pearson correlation coefficient between the ratio-based expression

levels of a dataset for a given pair of sample groups and the corresponding reference FC values (transformed to $\log_2$ scale). It is referred to as the 'relative correlation with reference datasets' metric, representing the numerical consistency of the ratio-based expression profiles. To improve reliability, calculate the mean of the three replicates of each sample group before performing ratio-based expression analysis. Run the script:

```
Rscript path-to/qc_rc.R -i path-to/example_log2_limma_data.csv -r path-to/example_ref_data.csv -o path-to/
```

▲ **CRITICAL** A file named `RC_each_batch_(date).csv` will be generated upon successful script implementation. This file contains the RC values between the test and reference datasets for each batch.

91. Calculate the Matthews correlation coefficient (MCC) of reference DEGs. MCC measures the consistency of DEGs detected from a dataset for a given pair of samples with those from the reference DEGs or 'MCC of DEGs' as follows:

- Repeat Step 68 to identify DEGs and non-DEGs of a test dataset
- Obtain reference DEGs and non-DEGs from Steps 69. They can be true positive (TP) and true negative (TN) sets
- Compare DEGs and non-DEGs with reference DEGs and non-DEGs. Compute the number of TP, TN, FP and FN
- Calculate the MCC using TP, TN, FP and FN values

Run the script:

```
Rscript path-to/qc_mcc.R -i path-to/example_log2_limma_data.csv -r path-to/example_ref_data.csv -o path-to/
```

▲ **CRITICAL** A file named `MCC_each_batch_(date).csv` will be generated upon successful script implementation. This file contains the TP, TN, FP, FN and MCC values between the test datasets and reference datasets for each batch.

92. Calculate the root mean square error (r.m.s.e.), using FCs between a test dataset for a given pair of samples and the corresponding ratio-based reference datasets, representing the average distances of ratio-based expression profiles. FCs were transformed using $\log_2$ scaling. Use `rmse()` function Metrics package⁴⁵ for implementation. Run the script:

```
Rscript path-to/qc_rmse.R -i path-to/example_log2_limma_data.csv -r path-to/example_ref_data.csv -o path-to/
```

▲ **CRITICAL** A file named `RMSE_each_batch_(date).csv` will be generated upon successful script implementation. This file contains the r.m.s.e. value between test datasets and reference datasets for each batch.

93. Compute criteria of the above three metrics, using the mean minus the standard deviation of RC and MCC and the mean plus the standard deviation of r.m.s.e. In our study, the criteria of RC, MCC and r.m.s.e. were set to 0.89, 0.54 and 0.38, respectively.

Proficiency test

● TIMING varies

▲ **CRITICAL** Laboratory proficiency can be monitored through a number of approaches through using RNA reference materials and datasets.

94. Distribute at least three units of RNA reference materials to different laboratories.

▲ **CAUTION** At least two or more different types of RNA reference materials shall be employed in proficiency testing to ensure that the ability to detect differential gene expression between sample types can be assessed.

95. Conduct RNA-seq experiments in different laboratories and generate raw sequencing data in fastq format. Repeat RNA-seq data analysis described in Steps 31–39.
96. Repeat Step 58 to assess quality using reference-dataset-free metrics, considering sequencing reads, alignment and expression profiling. Additionally, repeat Steps 90–92 for quality measurement based on reference-dataset-based metrics, including RC, MCC and r.m.s.e. Exclude poor-quality data from further analysis.

Diagnosis and correction of batch effects

● TIMING 2 h

97. Apply PCA and SNR to visually and numerically assess the ability to distinguish the intrinsic biological differences among the reference material sample groups from variations among technical replicates of the same sample group in terms of cross-batch integration. Repeat Step 56.
98. Use the principal variance component analysis (PVCA) method to assess whether batch effects are present in the dataset. When combined with bar plots, PVCA can quantify and visualize the proportion of variations attributed to experimental effects, including batch effects. Run the script:

```
Rscript path-to/pvca.R -i path-to/example_expr_multibatch_log2.csv -m path-to/metadata.csv -o path-to/
```

▲ **CAUTION** PVCA installation requires more dependent packages and may conflict with other R package versions. Creating a conda environment dedicated to PVCA analysis is recommended.

▲ **CRITICAL** Two files will be obtained after successfully implementing the script, including (1) the PVCA value of the expression profiles of multiple batches (RNA_PVCA_(date).csv) and (2) a bar plot of the PVCA results of the expression profiles of multiple batches (RNA_PVCA_plot_(date).pdf).

99. A reference-material-based ratio method should be applied for batch-effect correction if batch effects are detected or suspected. Specifically, ratio-based expression data are obtained within each batch on a gene-by-gene basis. For each gene, calculate the mean of expression profiles (for example, $\log_2$ FPKM) of replicates of the reference sample(s) and then subtract it from the $\log_2$ FPKM values of that gene in each study. Run the script:

```
Rscript path-to/ratio.R -i path-to/example_expr_multibatch_log2.csv -m path-to/metadata.csv -o path-to/
```

▲ **CRITICAL** Samples used as denominators for performing ratio-based methods shall be annotated in the metadata.csv file.

▲ **CRITICAL** This step is not obligatory and depends on the preceding diagnostic results. We recommend using $\log_2$ -transformed FPKM data in this step. A file named RNA_expression_ratio_(refsample)_(date).csv will be obtained after successfully implementing the script, which contains the ratio-based expression profiles of each gene.

100. After applying batch-effect correction, reevaluate the data using the SNR and PVCA methods to assess whether the batch effects have been successfully eliminated or reduced. Repeat Steps 97–98.

◆ TROUBLESHOOTING

Troubleshooting

Troubleshooting advice can be found in Table 3

Table 3 | Troubleshooting table

Step	Problem	Possible reason	Solution
56	Low SNR value	Poor data quality	Flag the data as poor quality
		'Random failure' of a particular technical replicate	Remove the 'failure' sample
		Sample label assignment error	Check in advance that the data labels and sample names are matched
58	Failure of sequencing data quality, including insufficient sequencing volume, insufficient sequencing read length and too many splice sequences	There is a problem with the sequencing site	Resequencing
	Presence of other species in the sequencing data or a large number of sequences of rRNAs and mtRNAs	Contamination with other RNAs introduced during library construction or sequencing	Reconstruct libraries and perform resequencing
	Low mapping ratio of the reference genome for study subjects	Sequencing file input error	Check sequencing files for correct grouping
	The gene body is unevenly distributed, or the genomic regions are not aligned as expected	Reference genome selection error	Selection of a suitable reference genome
66	Expression data are not normally distributed	RNA degradation during sequencing	Reconstruct libraries and perform resequencing
70	Expression data are not normally distributed	Expression data not suitably normalized	Try different normalization methods
70	A small number of detected DEGs	It may be right when the different groups of reference materials are biologically similar	No solution needed
88	A low number of genes that can be validated	The orthogonal technique was poorly chosen or executed	Ensure that the orthogonal technique is correctly executed, or change another orthogonal technique for validation
		The genes identified in the former part are FPs	Reassess the differential expression methods or change other aspects of the protocol
100	Batch effect still exists	Batch effect correction method was poorly chosen or executed	Ensure that the batch effect correction method is correctly executed or try different methods for correcting batch effects

Timing

Material preparation

Steps 1–15, purification of total RNA: 3 h

Steps 16–19, quality assessment of RNA samples: 1–2 h

Steps 20–23, subdividing RNA samples to aliquots: 1–3 h

Steps 24–26, quality assessment of RNA aliquots based on multiple laboratories: 2 h to 2 d

Homogeneity testing (optional)

Steps 27–28, sample selection: 1 h

Steps 29–30, library preparation and sequencing: 3 d

Steps 31–39, alignment and gene quantification: 6 h

Steps 35–39, quality control analysis: 2 h

Steps 40–42, compare variation between within-unit and between-unit groups: 1 h

Stability testing (optional)

Steps 43–44, short-term stability: varies

Steps 45–46, stability testing based on RIN: 4 h

Steps 47–48, stability testing based on RNA-seq: varies

Steps 49–52, long-term stability: varies

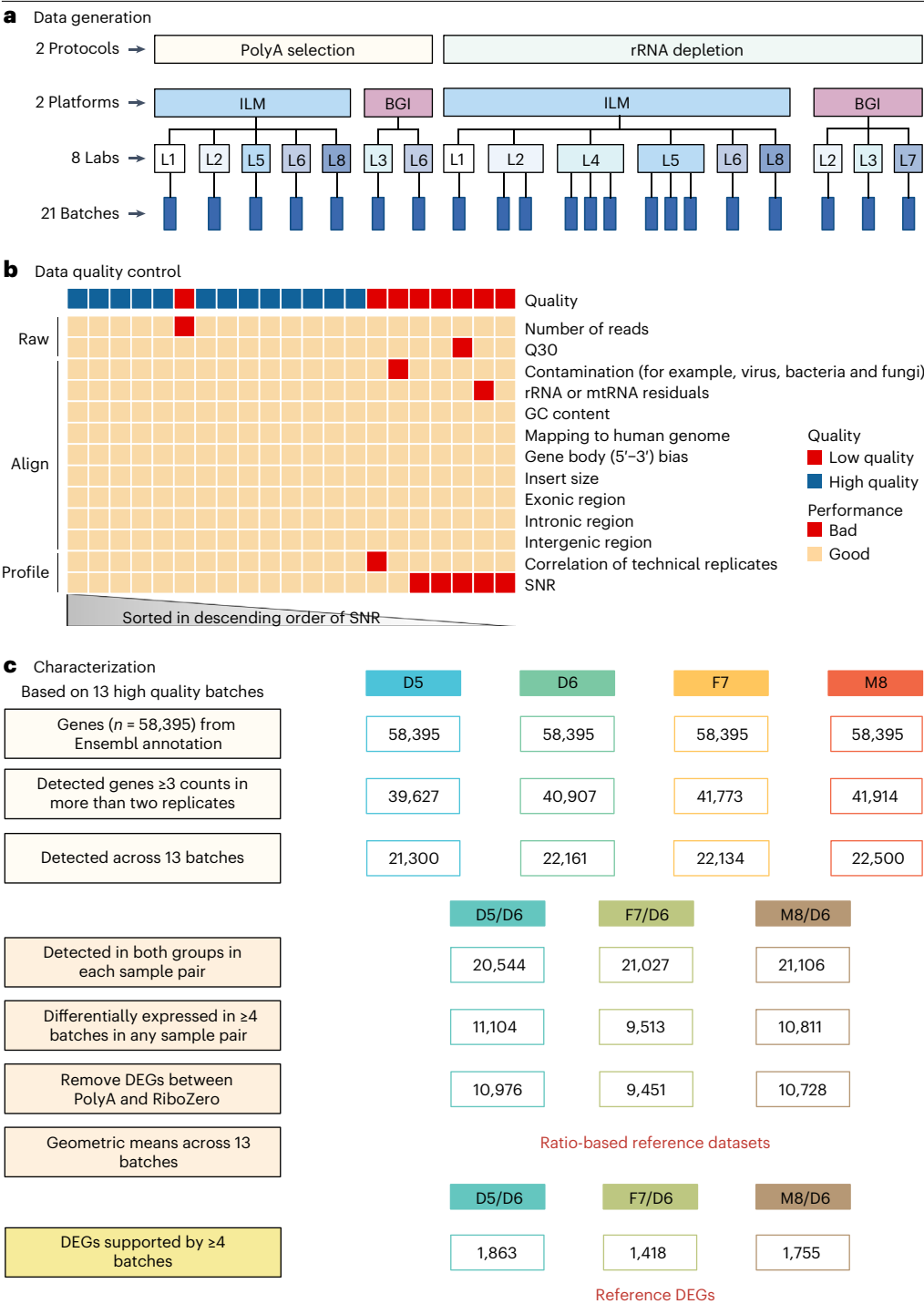


Fig. 5 | Characterization of Quartet RNA reference materials. **a**, Overview of the data generation of the Quartet RNA reference materials. BGI, MGIDNBSEQ-T7 sequencing platform; ILM, Illumina NovaSeq sequencing platform. **b**, Quality flags of RNA-seq batches, in terms of the number of sequencing reads, read quality (Q30, a quality score indicating that the probability of an incorrect base call is 1 in 1,000, corresponding to a base call accuracy of 99.9%), percentage of reads that were mapped to contamination species (for example, virus, bacteria and fungi), percentage of reads that were mapped to rRNA or mtRNA residuals, percentage of reads that were mapped to the human genome, gene body (5'-3') bias, insert size, percentage of mapped reads that were located in the exonic, intronic and intergenic region of human genome, Pearson correlation coefficient of technical replicates and SNR. The batches were ordered by SNR values. **c**, Workflow for constructing Quartet RNA reference datasets. Boxes are color-coded by sample ID.

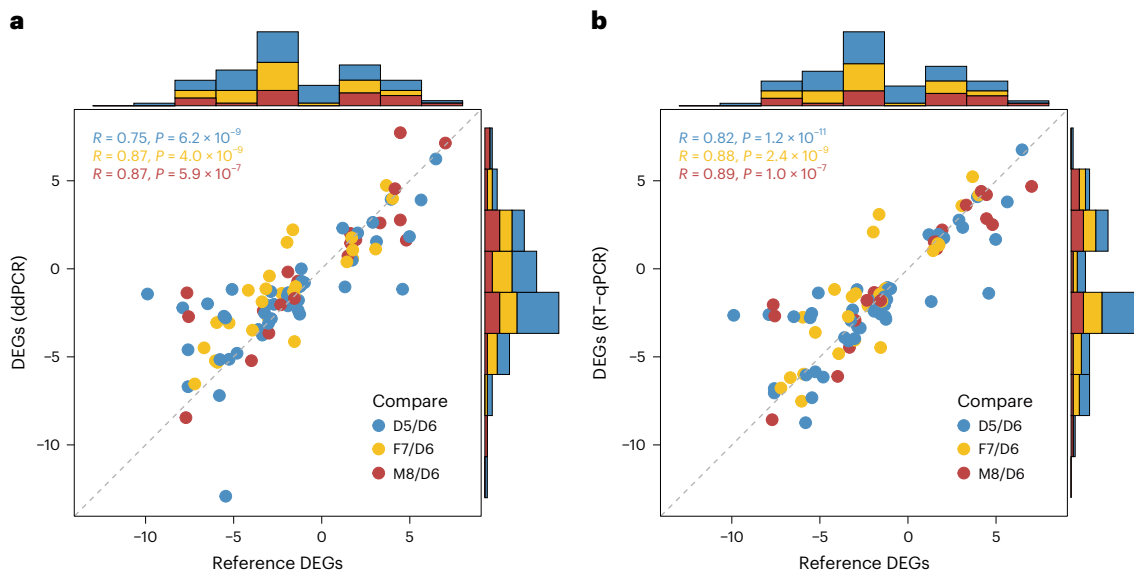


Fig. 6 | Validation of ratio-based transcriptome-wide reference datasets.

a,b, Scatter plots of $\log_2$ FCs of gene expression between reference DEGs and RT-qPCR (**a**) and ddPCR (**b**). Pearson correlation coefficient across three sample pairs was calculated. The genes that were considered differentially expressed in both methods, shown in the x axis and y axis, were used for plotting.

x axis: average $\log_2$ FC from 13 high-quality RNA-seq batches' reference DEGs.
 y axis: for the RT-qPCR, a gene was considered as a DEG when the *t*-test two-sided $P < 0.05$ and $FC \geq 2$ or ≤ 0.5 ; for ddPCR data, the genes that were identified as DEGs based on RT-qPCR were used. Average $\log_2$ FC of RT-qPCR ($n = 3$) and ddPCR ($n = 2$) data from DEGs were used for plotting.

Characterization

Steps 53–54, RNA-seq from multiple protocols, platforms, laboratories and replicates: varies

Steps 55–57, data quality control based on expression profiles: 2 h

Step 58, identification of high-quality RNA-seq datasets: 0.5 h

Steps 59–61, identification of high-confidence detected genes: 0.5 h

Steps 62–67, characterization of reference datasets at ratio level: 0.5 h

Steps 68–70, identification of reference DEGs: 0.5 h

Uncertainty estimation

Steps 71–76, measurement uncertainty estimation: 2 h

Orthogonal validation

Steps 77–81, RT-qPCR: 3 h

Steps 82–86, ddPCR: 3 h

Steps 87–88, validation assessment: 0.5 h

Performance metrics

Steps 89–93, construction of performance metrics: 2 h

Proficiency test

Steps 94–96, proficiency test: varies

Diagnosis and correction of batch effects

Steps 97–100, diagnosis and correction of batch effects: 2 h

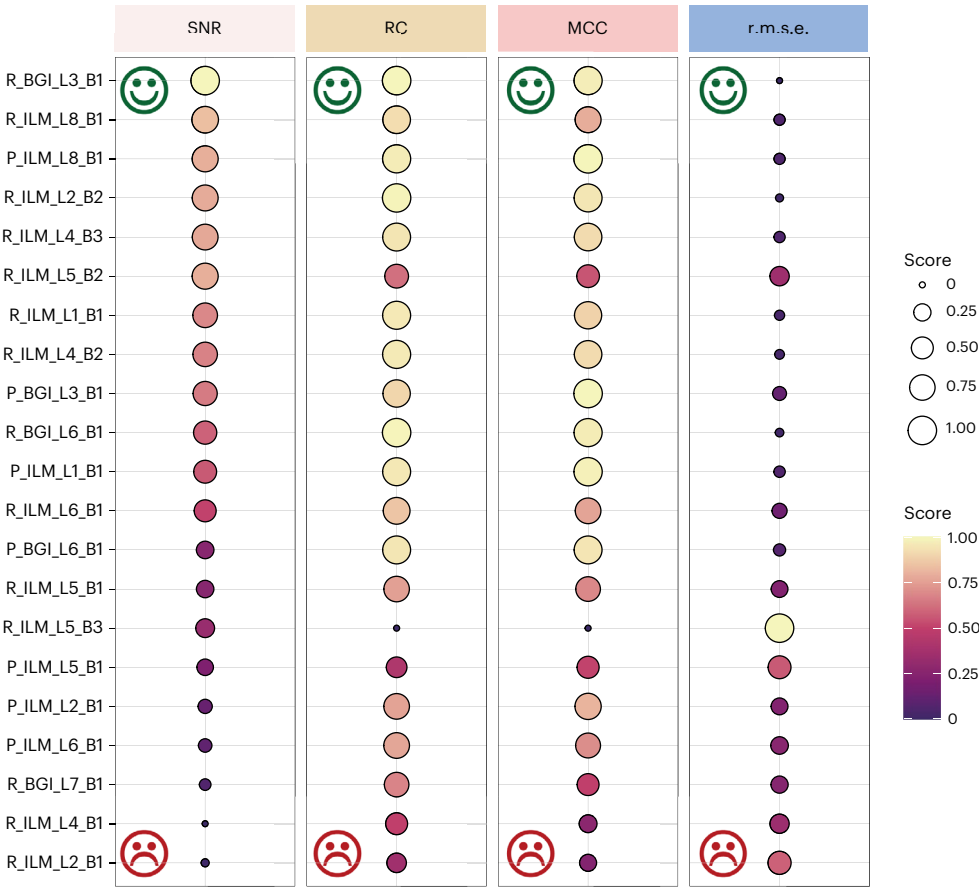


Fig. 7 | Summary of different quality-control metrics based on reference datasets. This summarizes the performance of SNR, RC, MCC and r.m.s.e. in 21 RNA-seq batches. To equalize the weight of different metrics, the values of the four metrics were scaled to an interval of [0,1]. Happy faces represent high performance; sad faces represent low performance.

Anticipated results

We anticipate that establishing RNA reference materials and reference datasets will greatly enhance the quality of transcriptomic data for various scientific and translational applications.

Using the development of the Quartet RNA reference materials and reference datasets as an example, we obtained 252 RNA-seq libraries from 21 batches generated in eight laboratories using two library construction protocols (poly(A) selection and RiboZero) and two sequencing platforms (Illumina NovaSeq and BGI DNBseq) (Fig. 5a). In the study, a batch is defined as 12 libraries from a standard sample set, consisting of 12 tubes with each representing one of the triplicates of the Quartet RNA reference sample groups, whose library construction and sequencing experiments were conducted simultaneously. Using multiple metrics, including SNR and other widely used quality metrics with fastq, bam and expression profiles, 13 batches were flagged as high quality. They were used for subsequent data integration to create the reference datasets (Fig. 5b,c). Furthermore, the reference DEGs were randomly selected to be validated using orthogonal techniques such as qPCR (Fig. 6a) and ddPCR (Fig. 6b).

Multiple performance metrics were then calculated on the basis of 21 batches, including SNR, RC, MCC and r.m.s.e. (Fig. 7). SNR was utilized to assess the capability of RNA-seq data in distinguishing between different groups of biological samples, with higher levels indicating

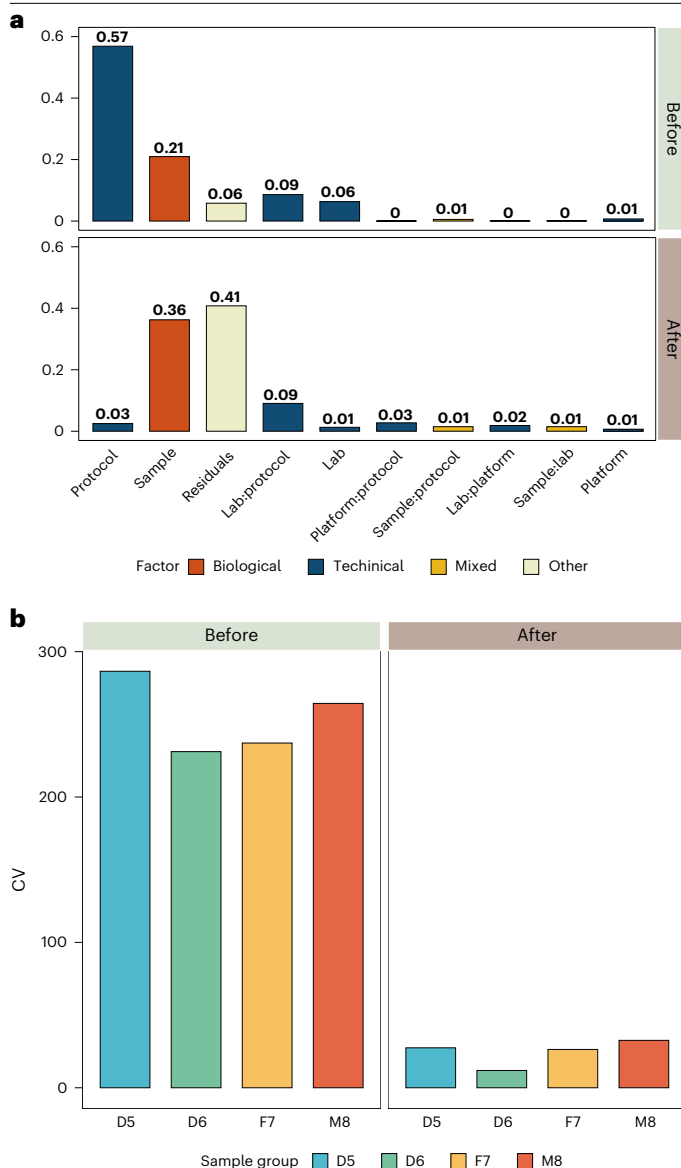


Fig. 8 | Correcting batch effects using a reference-material-based ratio method. Examples of visualization plots before and after batch correction. **a**, PVCA of the relative contribution of main effects (protocol, platform, lab and sample group) and their combinations (:) to total model variance. The y axis shows the proportion of variance attributed to each effect, while the x axis labels correspond to the different effects being analyzed. **b**, The coefficient of variation (CV) of the expression pattern of the representative gene across batches.

superior data quality. Furthermore, we employed RC and r.m.s.e. derived from reference datasets to evaluate gene-expression agreement and FC disparity between profiled samples and reference datasets, respectively. Higher RC values and smaller r.m.s.e. values indicate greater concordance in analysis results. Additionally, the MCC based on reference DEGs was used to evaluate the consistency between DEGs identified in test samples and the reference DEGs. A higher MCC value indicates a more consistent test result. We present the combined performance of these four metrics using the publicly available 21 batches of the Quartet RNA datasets (Fig. 7).

When using the RNA reference materials and reference datasets for diagnosing and correcting batch effects, we conducted PVCA to evaluate the factors influencing the quality of the Quartet RNA-seq datasets across 21 batches. The contributions of these factors were depicted through bar plots. It became apparent that in the dataset without batch correcting, the protocol factor (that is, polyA or RiboZero protocol) exerted the most substantial influence, indicating the presence of batch effects. However, upon implementing a reference-material-based ratio method, the sample factor, which represents the primary focus of the study,

emerged as the most significant contributor (Fig. 8a), signifying the successful correction of batch effects. Additionally, we randomly selected some genes with significant differences in gene expression profiles across different batches. After using the reference-material-based ratio method, these genes showed similar expression patterns across different batches (Fig. 8b).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The example RNA-seq sequence data and gene expression data have been deposited in the GSA (accession number: HRA001859)⁴⁰ and Open Archive for Miscellaneous Data (OMIX) (accession number: OMIX002254)⁴⁶ of China National Center for Bioinformation. Moreover, we have developed the Quartet Data Portal (<http://chinese-quartet.org>) for the community to access and share the Quartet multiomics resources¹⁹. The example data for multiple analyses, such as homogeneity testing, stability testing, characterization and performance assessment, are available at GitHub (https://github.com/YingYu12345/Reference_standard_protocol). Source data are provided with this paper.

Code availability

This protocol provides scripts for data analysis and visualization of the results generated in the ‘Procedure’ section at the GitHub repository at https://github.com/YingYu12345/Reference_standard_protocol and Zenodo repository at <https://doi.org/10.5281/zenodo.13777366> (ref. 41).

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Author contributions

Y.Z., L. Shi, L.D., R.Z. and Y.Y. conceived and supervised the study. Y.Z. and W.H. were responsible for the wet-lab implementation of the study. Y.Y., Q.C., W.H. and X.G. constructed figures and tables. Y.Y., Q.C., X.G., L. Sang, H.X., D.W., J.L., X.F., R.Z., L.D., L. Shi and Y.Z. performed data analysis and/or interpretation. Y.Y., Q.C., W.H., X.G., L. Sang, H.X., L. Shi and Y.Z. wrote and/or revised the manuscript. All authors reviewed and approved the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
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☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

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Software and code

Policy information about [availability of computer code](#)

Data collection	The details of data collection methods used were described in the Procedure section.
Data analysis	The GitHub (https://github.com/YingYu12345/Reference_standard_protocol) and Zenodo (https://zenodo.org/doi/10.5281/zenodo.13777366) repository of this protocol provides scripts for data analysis and visualization of the results generated in Procedure.

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The example data for multiple analyses, such as homogeneity testing, stability testing, characterization, and performance assessment, are available at GitHub (https://github.com/YingYu12345/Reference_standard_protocol).

Human research participants

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Reporting on sex and gender	Data were obtained from published paper (PMID: 37679545).
Population characteristics	NA
Recruitment	NA
Ethics oversight	NA

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

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All studies must disclose on these points even when the disclosure is negative.

Sample size	Data were obtained from published paper (PMID: 37679545).
Data exclusions	The example data were obtained from published paper (PMID: 37679545).
Replication	Data were obtained from published paper (PMID: 37679545). The reference materials were profiled within a batch in a lab in three replicates for each of the four sample groups (donors).
Randomization	NA
Blinding	NA

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☒	☐ Eukaryotic cell lines
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☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging